# Supplementary Methods

*Measurement contract, coordinate construction, and paired inference*

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## S1. Dataset and H5 measurement contract

The analysis used 68 H5 files representing 34 participants by two separately excerpted conditions: euglycemia and hypoglycemia. Each H5 file contained 74 overlapping windows from an approximately five-minute EEG segment. The contract included raw and robust-z EEG features, 3D MNPS, 9D coordinates, local 3D/9D Jacobians, and subject- and cohort-anchored coordinate layers. It did not include participant-level glucose, a shared time reference, recovery, regional MNPS, anchor-state, anchor-quality, anchor-coupling, or block-window exports.

## S2. Preprocessing and epochs

The embedded ingest configuration specified 19-channel scalp EEG at 200 Hz, 1–45 Hz bandpass filtering, a 50 Hz notch filter, and average rereferencing. Windows were 8 s with a 4 s step. The H5 provenance reported 300 s of measured coverage and 296 s of effective window coverage per condition.

## S3. Coordinate layers

The 9D coordinate order was $(m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m)$. The 3D export was a fixed weighted projection of these three coordinate families. In the cohort-anchored layer, feature centers and scales were fit from pooled euglycemia and hypoglycemia rows. This makes between-condition coordinates comparable within the released surface, but it does not make euglycemia an external or physiological zero.

Let $z_t$ be the 9D chart and $P$ its fixed projection. Then

$$x_t = Pz_t, \quad x_t = (m_t, d_t, e_t).$$

The projection is an analysis contract. It is not evidence that the three coordinates are a unique ontology of neural state.

## S4. Local flow and reachability

For a local linear approximation, the exported dynamics estimate a Jacobian $J_t$ and a discrete transition operator $A_t$. Reachability was summarized from

$$\Sigma_t^{h+1} = A_t \Sigma_t^h A_t^T + Q_t,$$

where $Q_t$ is a local residual covariance. We report tube log-determinant as a volume proxy, effective dimension, condition number, directional entropy, and a rotation proxy. These metrics describe local model-propagated geometry; they do not establish actual state occupancy, controllability, cognitive performance, or consciousness.

## S5. Primary paired inference

All analyses used participant summaries. For the engine-level analysis, euglycemia was passed as the left condition and hypoglycemia as the right condition because the implementation reports `right - left`. The contrast is therefore HYPO minus EU throughout the manuscript and tables.

The engine used 5,000 paired permutations, bootstrap confidence intervals, and Benjamini–Hochberg false-discovery-rate correction within contrast and analysis type. The primary inferential surface contained 3D/9D coordinate, 3D Jacobian, and reachability blocks. One unavailable 9D Jacobian reduced the testable reachability sample to 33 pairs.

## S6. Robustness protocol

The article-local robustness script performed a 10,000-draw paired label-swap null on 11 prespecified 9D and reachability medians. Each draw independently multiplied every participant's HYPO minus EU difference by

plus or minus one; the two-sided Monte Carlo probability compared the observed mean difference to the absolute null distribution. This is the appropriate label permutation for the paired two-condition design.

The script additionally read H5 geometry-contract provenance and excluded the three hypoglycemia sessions marked `adjusted`: `sub-004`, `sub-017`, and `sub-021`. It reports whether the direction of each endpoint is preserved. It does not rerun FDR-corrected restricted-sample inference, so this sensitivity is directional rather than a replacement primary test.

## S7. Feature-level sanity checks

Median raw values were extracted for delta, theta, alpha, sample entropy, permutation entropy, and Hjorth complexity. Paired Wilcoxon tests (Wilcoxon, 1945) and six-feature Benjamini–Hochberg correction were used for this secondary check. The exported sample- and permutation-entropy fields were identical for this H5 release; their separate table rows are therefore not independent replications.

## Bibliography

Wilcoxon, F. (1945). Individual Comparisons by Ranking Methods. *Biometrics Bulletin*, *1*(6), 80–83. https://doi.org/10.2307/3001968